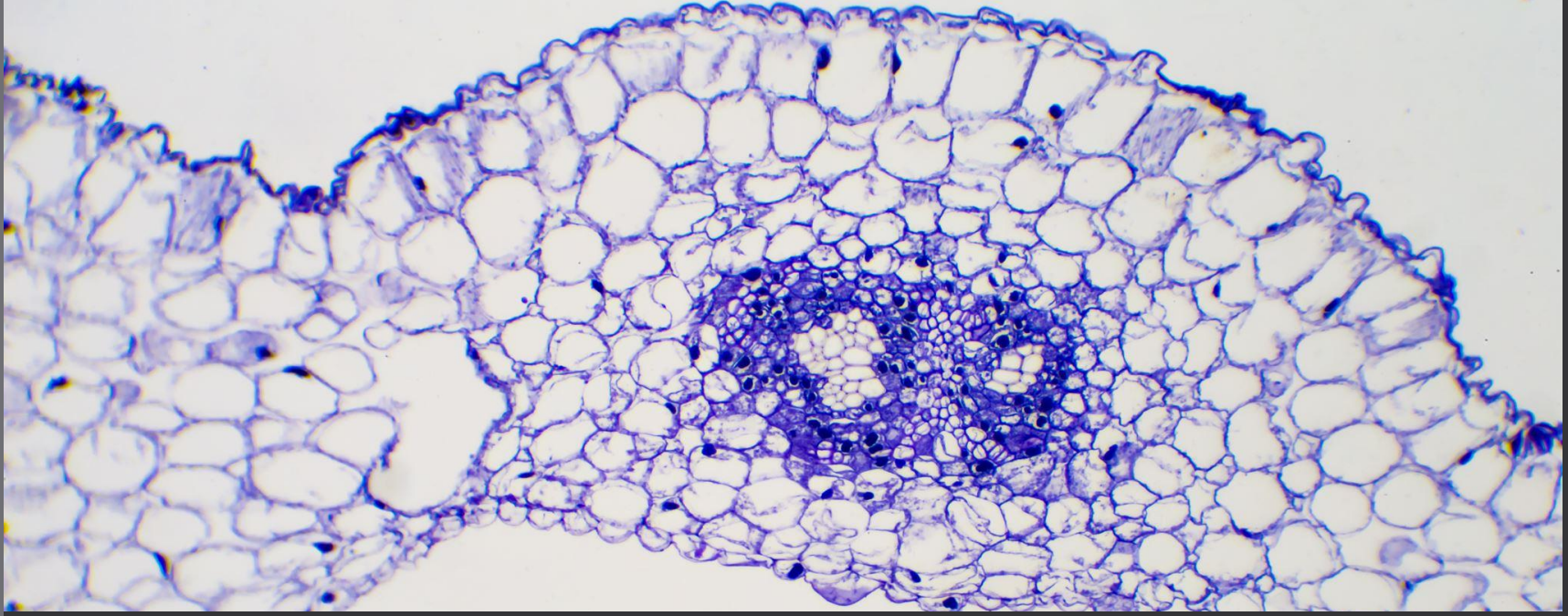
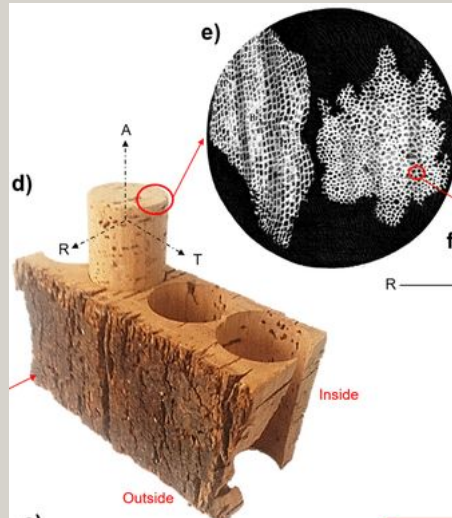
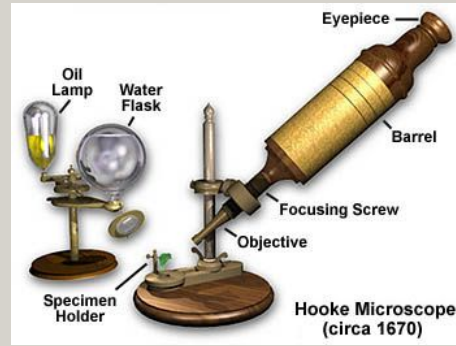




Unit 2 (Part 2): Cell Theory and Cell Structure

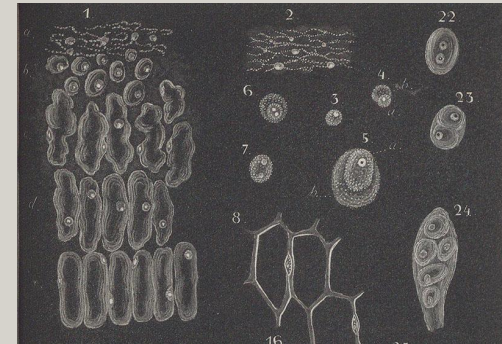
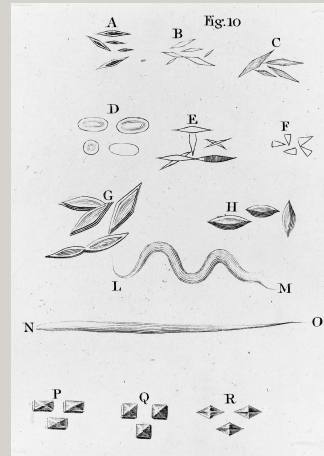
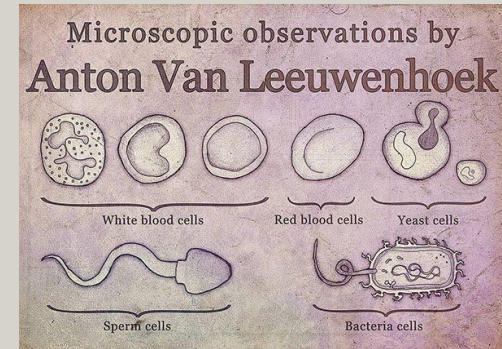
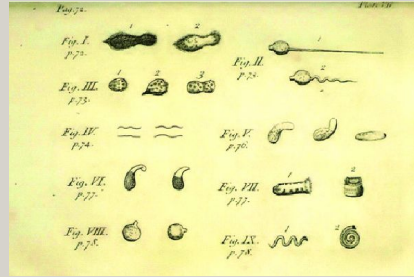
Cell History

- **Robert Hooke** - used primitive microscope to look at **cork** (in 1665!)
- Saw thousands of tiny chambers which **he called “cells”** (they were actually dead plant cells)



Cell History Cont.

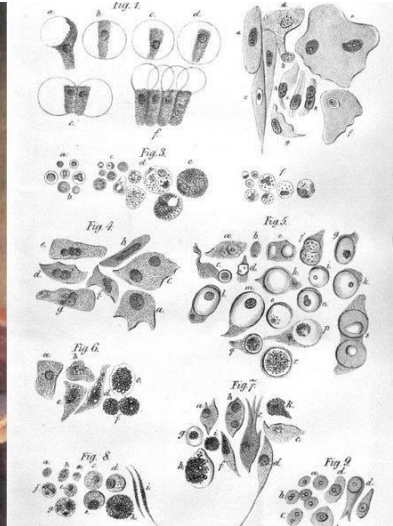
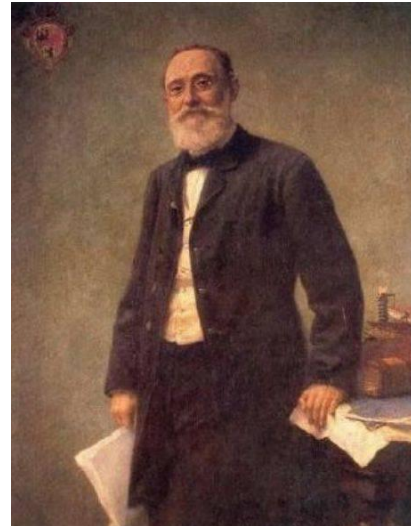
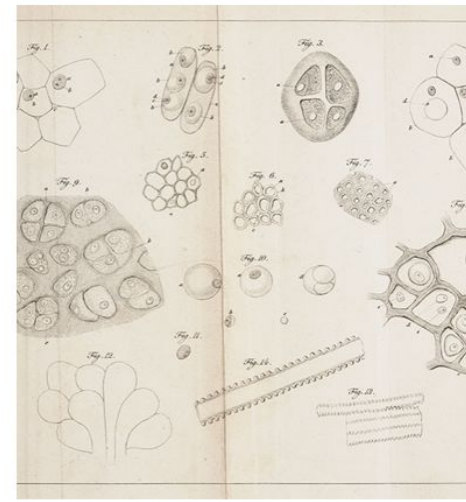
- Anton Van **Leeuwenhoek**:
 - First to observe microorganisms
 - Called them “**animalcules**” (“little animals”)
- Matthias **Schleiden**:
 - Proposed that **all plants are made of cells**



Cell History Cont.

- **Theodor Schwann:** proposed that all animals are made of cells (“swan”)
- **Rudolf Virchow:** proposed that all cells come from pre-existing cells
- All of these scientists contributed to what is known as **cell theory**

Theodore Schwann

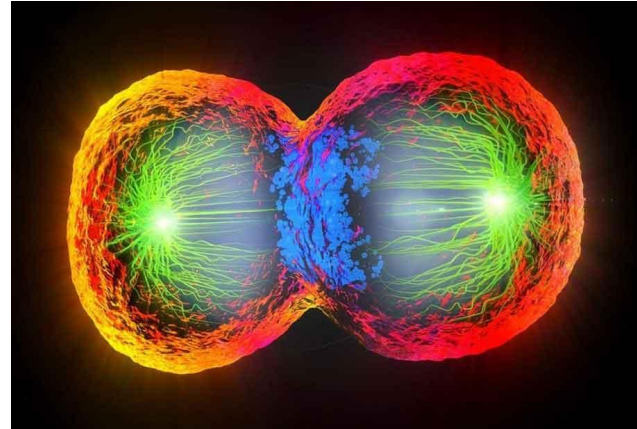


3 Parts of Cell Theory

All living things are made of **cells**

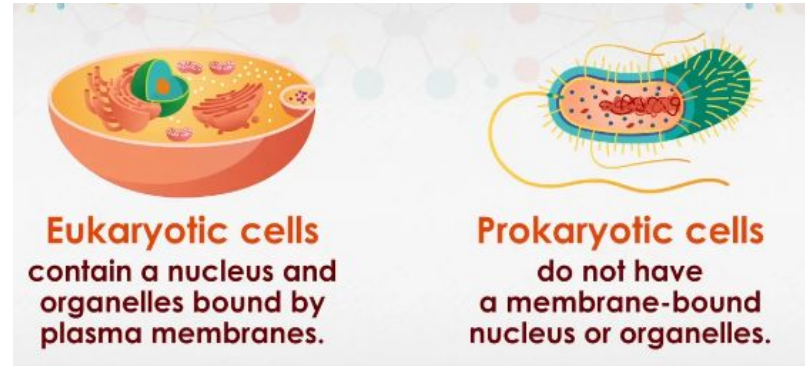
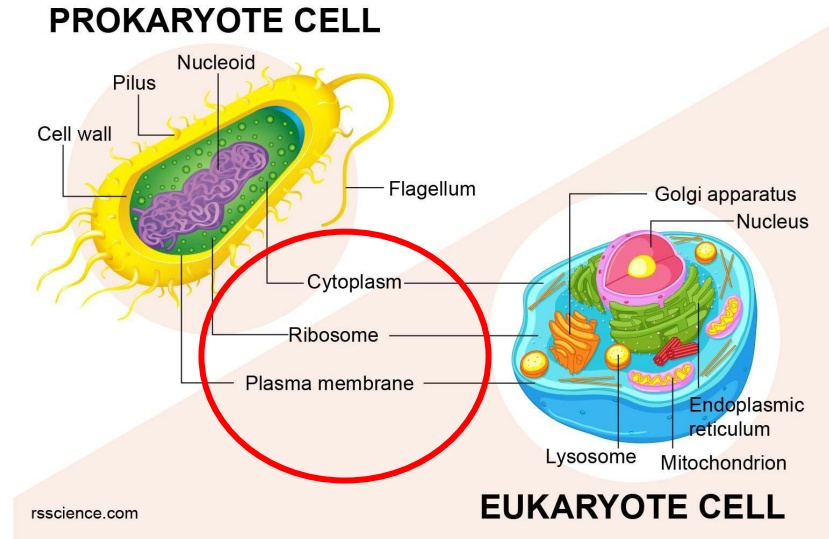
Cells are the **basic unit of life**

All cells come from **other cells**



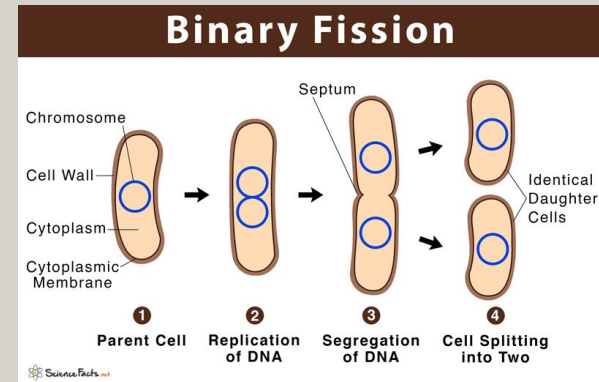
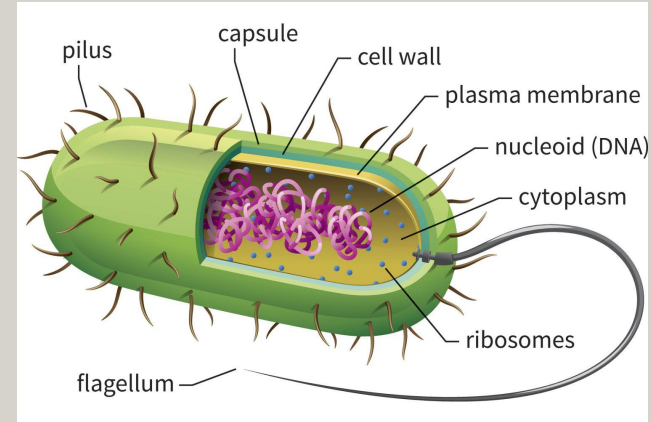
2 Types of Cells

- **Eukaryotes**: plants, animals, protists, and fungi
- **Prokaryotes**: **bacteria** (and Archaea, but we'll cover them later)
- ****ALL CELLS HAVE****:
 - Cytoplasm
 - Ribosomes- make proteins (**not considered an organelle**)
 - Cell membrane
 - Genetic Material (DNA or RNA)



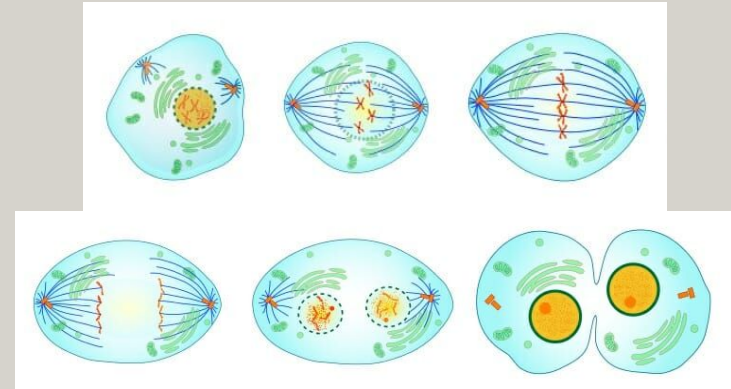
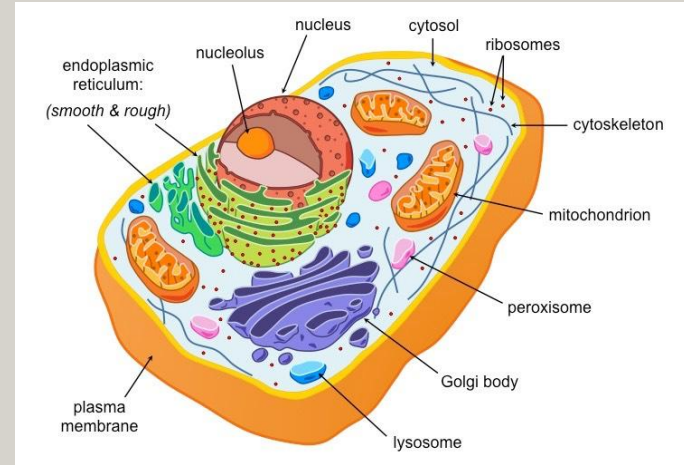
Prokaryotic Cells (Bacteria & Archaea)

- All **unicellular organisms**
- Smaller than eukaryotic: 1-5 micrometers
- **No nucleus OR membrane-bound organelles** (“pro” rhymes with “no”)
- “Nucleoid” region contains **circular DNA**
- Surrounded by a **cell wall** (made of **peptidoglycan**) and **capsule**
- Smaller ribosomes than eukaryotic (70s)
- Reproduces by **binary fission**
- May have flagella to move



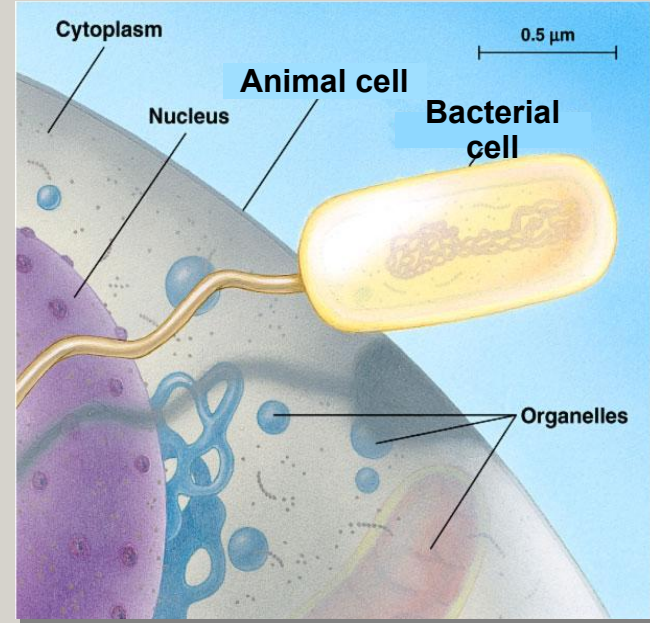
Eukaryotic Cells

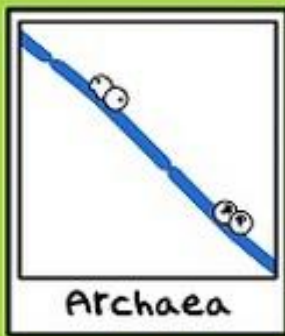
- Can either be **unicellular or multicellular**
- Plants, animals, protists, and fungi
- Larger size- 10-100 micrometers
- Have a **nucleus with linear DNA** and nucleolus - **“eu” = “true nucleus”**
- **Plant and fungi cells:** cell wall
- Have **membrane-bound organelles**
- Larger ribosomes (80s)
- Reproduce using **mitosis**



Cell Size Comparison

- Most bacteria: 1-10 micrometers (= 0.001 mm - 0.01 mm)
- Most eukaryotic cells: 10-100 micrometers (= 0.01 mm - 0.1 mm)
- Average human hair is about 0.016 mm to 0.05 mm thick

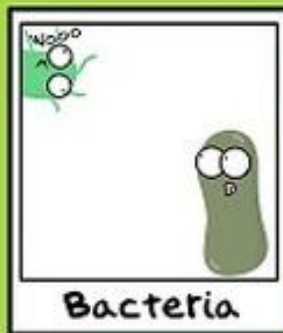




Archaea

PROKARYOTES

Pro- = "Before" Karyo- = "Nucleus"

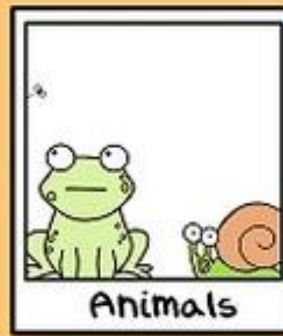


Bacteria

#AmoebaGIFs
@AmoebaSisters



Fungi

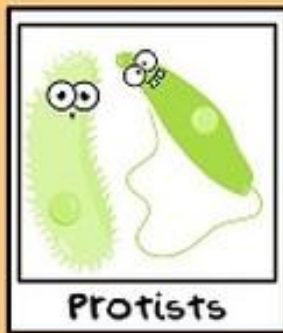


Animals

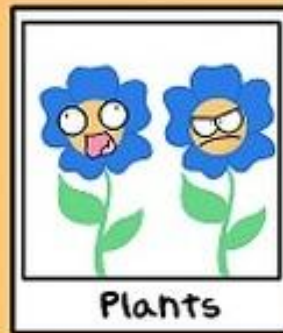
VS

Eu- = "True" Karyo- = "Nucleus"

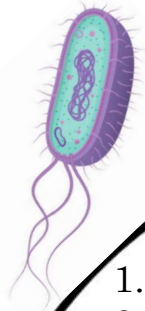
EUKARYOTES



Protists



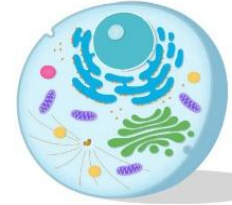
Plants



Prokaryotes

1. **NO** nucleus
2. **NO** membrane-bound **organelles**
3. **Smaller** in size (1-10 micrometers)
4. **Circular DNA** (plasmids)
5. **Unicellular ONLY**
6. Reproduce via **binary fission** ("b" for "bacteria")
7. **Cell walls** made of **peptidoglycan**

Eukaryotes

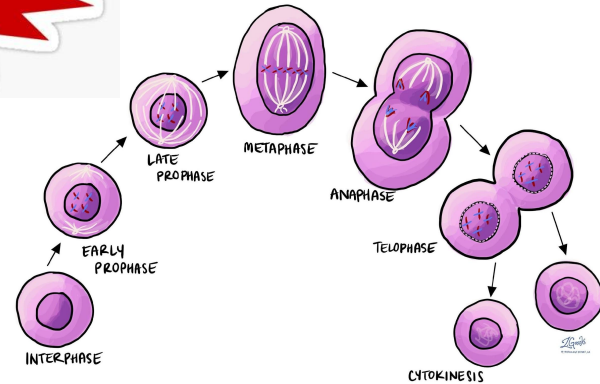
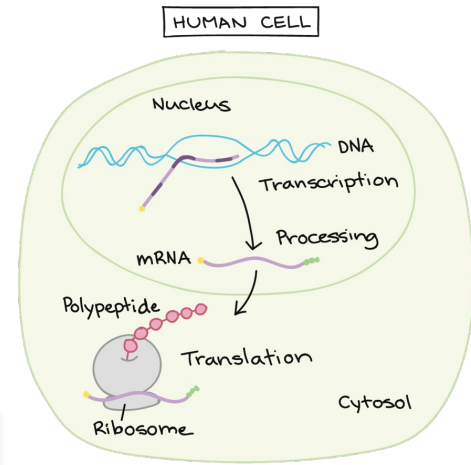
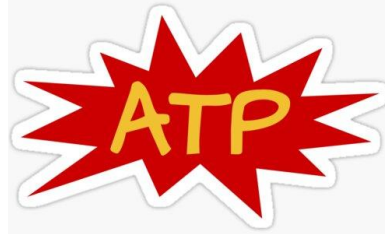


1. **YES** nucleus
2. **YES** membrane-bound **organelles**
3. **Larger** in size (10-100 micrometers) and more **complex**
4. **Linear DNA**
5. **Unicellular** (some protists & fungi) **OR** **multicellular**
6. Reproduce via **mitosis**
7. **Cell walls** made of **cellulose** (plants only) or **chitin** (fungi only)

1. **YES** cytoplasm
2. **YES** ribosomes (not a membrane-bound organelle)
3. **YES** cell membrane
4. **YES** genetic material (DNA or RNA)

3 Main Jobs of Cells

1. Make **energy**: need energy for all activities
2. Make **proteins**: do all the work in cells
3. Make **more cells**: growth and replacement



Eukaryotic Cell Comparison

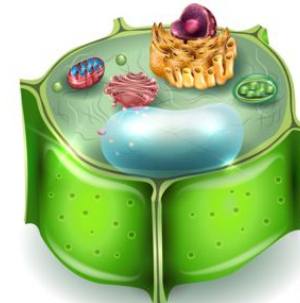
Animal Cells

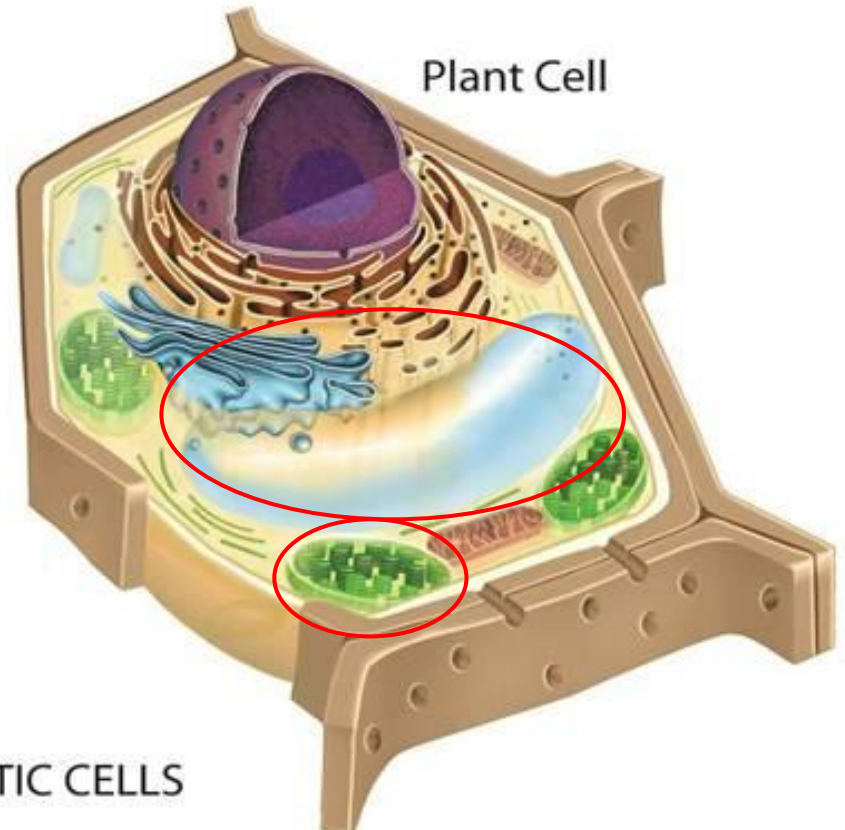
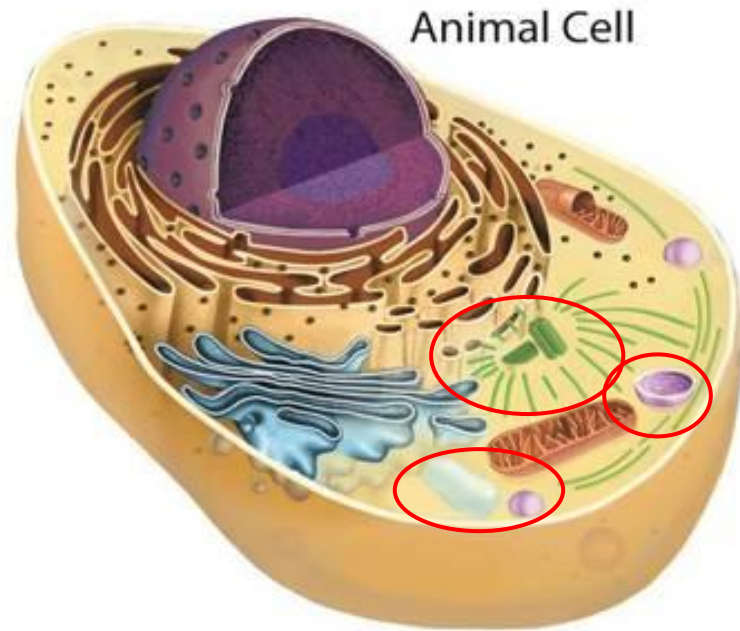
- **No cell wall**
- Irregular shape, less rigid
- **Many small vacuoles**
- Specialized organelles: **lysosomes** (digestion & “cleaning”) and **centrioles** (reproduction)



Plant Cells

- Have cell walls made of cellulose
- Cell wall causes fixed shape
- **ONE LARGE central vacuole** (storing water, food, waste, etc.)
- Specialized organelle: **chloroplasts** (give green color- **convert sunlight to food** for the cell)



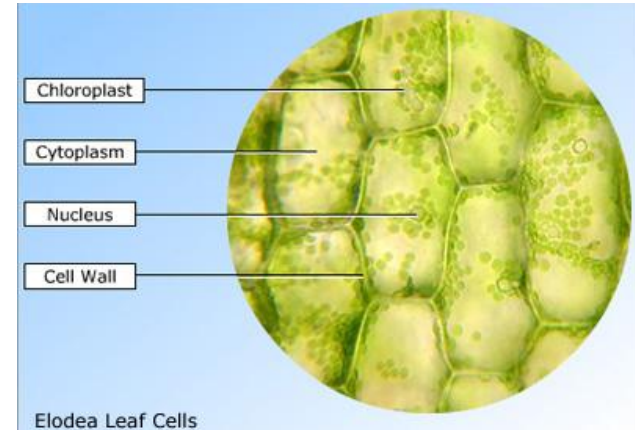


EUKARYOTIC CELLS

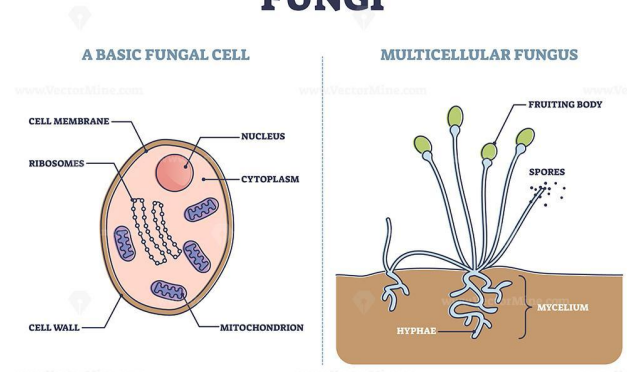
Eukaryotic Cells: Cell Walls (detailed)

- **Plants and Fungi ONLY**
- Provides **structure and support**
- Found **OUTSIDE** the membrane
- Layers of complex carbohydrates
 - **Cellulose** in **plants*****
 - **Chitin** in **fungi*****

NOT PRESENT IN ANIMALS



FUNGI



Eukaryotic Cells: Vacuoles (detailed)

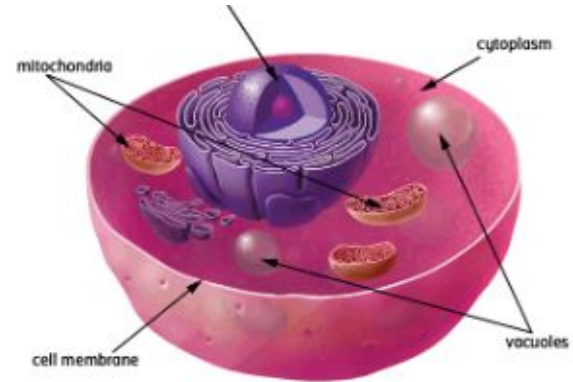
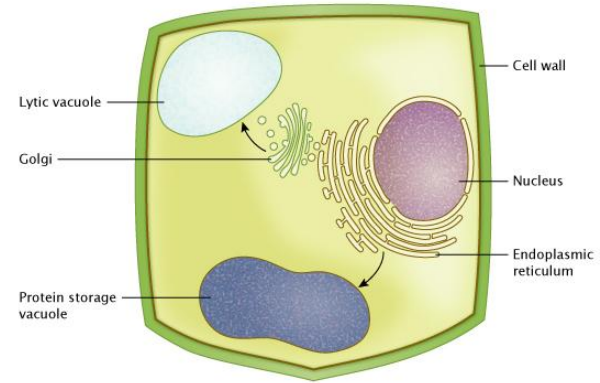
• Plants

- **One LARGE central vacuole**

- Holds water & helps prevent bursting in hypotonic environments
- Helps **give shape** to cells

• Animals

- Lots of **small vacuoles**
- May store **nutrients**
- May help with movement in protists (**contractile vacuoles** also push out excess water in hypotonic environments)



Animal Cell Surfaces

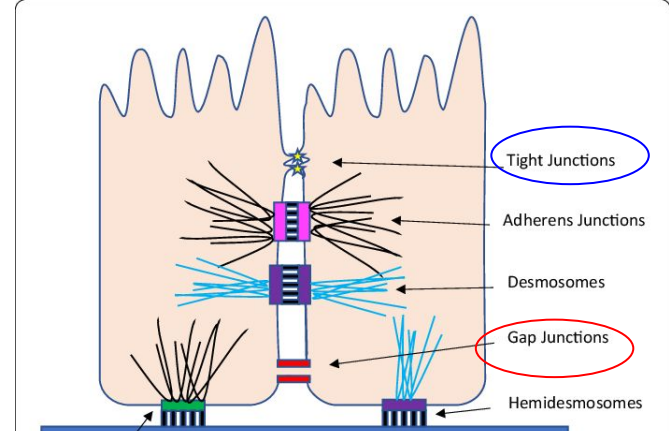
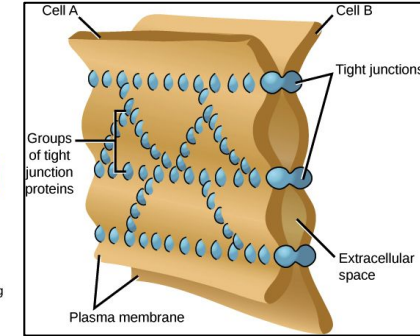
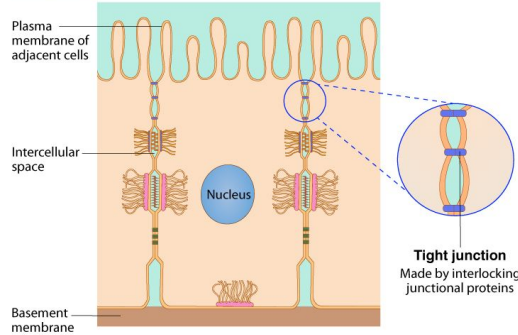
Tight Junctions: Made by interlocking proteins

- helps hold cells together

Gap Junctions: Directly connect cytoplasm of two cells

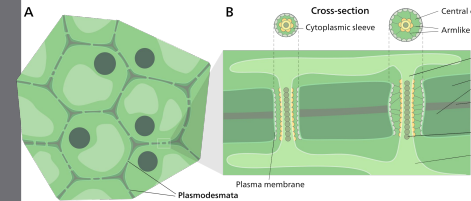
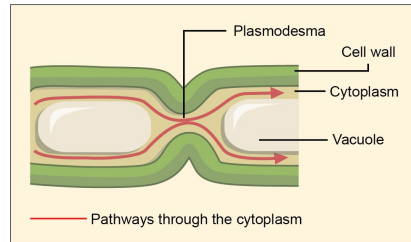
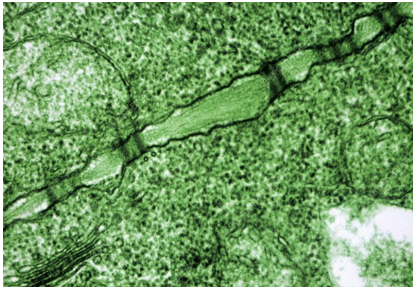
- Allows direct passage of molecules, ions, and electrical impulses through a gate between cells

Tight junction

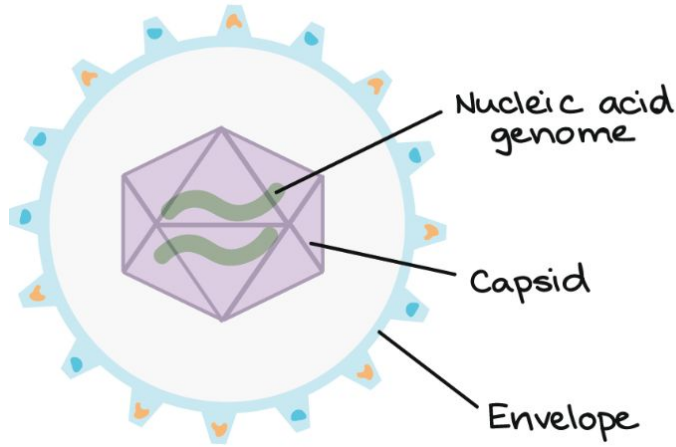


Plant Cell Surfaces

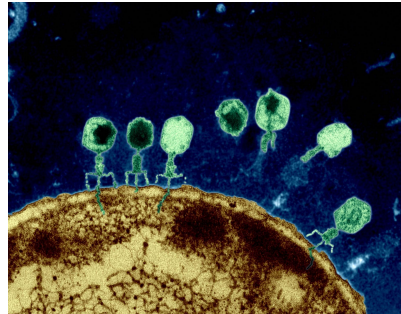
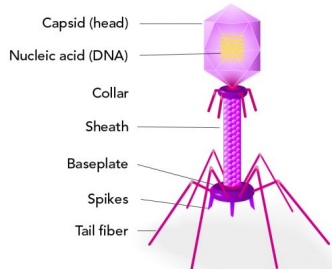
- Plasmodesmata- pores in the cell wall that interconnect plant cells
- Allows for cytoplasmic exchange, transport of substances and cell-to-cell communication within the plant since they are immobile (cannot move)



Viruses



Structure of a bacteriophage



- “Infectious particles”
- Have some BUT NOT ALL characteristics life
 - Needs a host to survive, no cells, no energy use
 - Can evolve RAPIDLY
 - Have genetic variation (“strains”)
- **Much smaller** than eukaryotic and prokaryotic cells
- All viruses have:
 - A **capsid**- protective shell
 - **Nucleic Acid Genome**-made of DNA or RNA
 - **Envelope**- layer of membrane
- Viruses “reproduce” by hijacking a cell’s machinery